Table 1. Data and analysis sets and video-derived representations.

(A) Data and analysis sets

Set	Cases	Cases per centre (2 / 3 / 4 / 5)	Definition	Design cells	Arms run[a]
Representation set[b]	278	49 / 41 / 71 / 117	279 counted cases minus one case (timestamp beyond video end); centre 1 excluded (mostly 5-min clips)	—	—
Primary causal set	158	28 / 34 / 44 / 52	s − 45 ≥ 0 and s + 60 ≤ n (n = floor(duration / 5)); offsets O = −15…15 (seed 20261005)	240	12
Per-W set W = 0	204	37 / 40 / 57 / 70	primary offsets kept; 46 new offsets, same rule and seed	54	6
Per-W set W = 5	201	37 / 40 / 56 / 68	primary offsets kept; 43 new offsets	6	6
Single-file set	123	13 / 34 / 44 / 32	primary cases with one video file	66	4
Excluded from the primary set	120	21 / 7 / 27 / 65	mainly early switches (46 of 117 centre-5 cases show the appendix within 75 s)	—	—

(B) Representations (models and checkpoints)

Arm[c]	Model and checkpoint	Kind	Input	Representation	Reduction
FM-P (primary, standardized)	SurgeNetCholec (CAFormer-S18) TimJaspersTue/SurgeNetModels @cc21f4a1 CHOLEC_checkpoint_epoch0050_ teacher.pth	frame-level	2-fps grid frames; crop, mask, 224², ImageNet normalisation	pooled normalised features, 512-d; mean of the step's 10 grid frames	PCA 64 (+ LEACE for FM-P−C)
FM-P-nat (native sensitivity)	SurgeNetCholec (as above)	frame-level	2-fps grid frames; resize only	as FM-P	PCA 64 (no −C arm)
FM-S (secondary)	LeanVAE-4ch Yumic/LeanVAE @4585e461 LeanVAE-dim4.ckpt	clip-level (L = 17 grid frames)	2-fps grid frames; crop, mask, 256²	posterior mean 4 × 32 × 32 averaged over latent time, 4,096-d	PCA 64 (+ LEACE for FM-S−C)
REF (reference)	nvidia/Cosmos-1.0-Tokenizer-CV8x8x8 @5110dbb3 (diffusers AutoencoderKLCosmos)	native clip (≤ 121 native frames per 5-s step)	crop, mask, 320²	post_quant_conv(latent mode) averaged over latent time, 16 × 40 × 40 = 25,600	PCA 64 via randomized SVD (+ LEACE for REF−C)

Cases: full-length DS203 videos of centres 2–5. Design cells: grid cells run in the set, each with 200 synthetic replicates. [a]Arms run (Figure 1C): primary causal set, all 12 arms; per-W sets W = 0 and W = 5, FM-P, FM-P−C, NV, ORACLE, ORACLE−C, SITE; single-file set, FM-P, FM-P−C, NV, SITE.

[b]E1, E3-site, principal component analysis (PCA) and LEACE fitting; no causal grid. [c]Selection (protocol §4.1 ranking and tie-breaks; Decision 005 for the reference): FM-P, §4.1 rank 1 (after CODELOCK_CORRECTION_001); FM-P-nat, §3.2; FM-S, §4.1 secondary rank 1 (tie-break 2: GFLOPs); REF, Decision 005 (not ranked). s, switch step of the content state; n, number of 5-s steps; O, window offset; −C, linearly site-removed; nat, native frames.